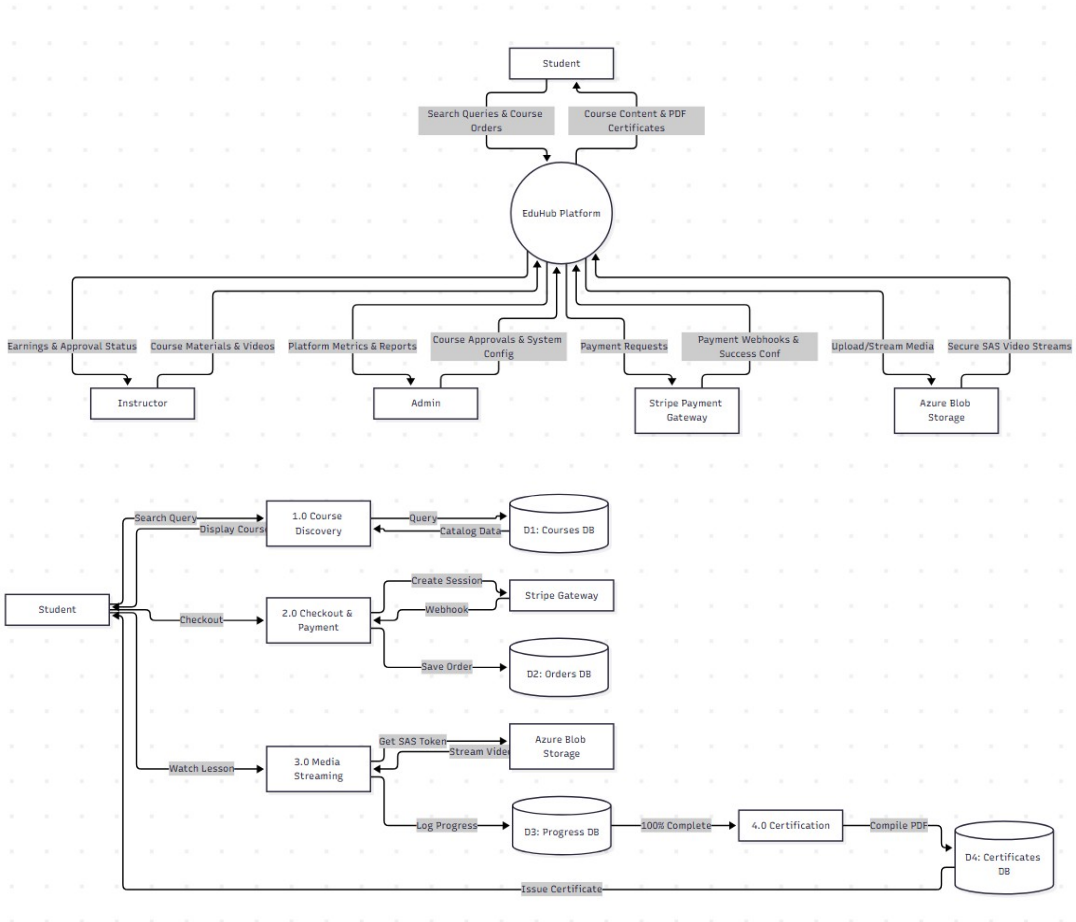
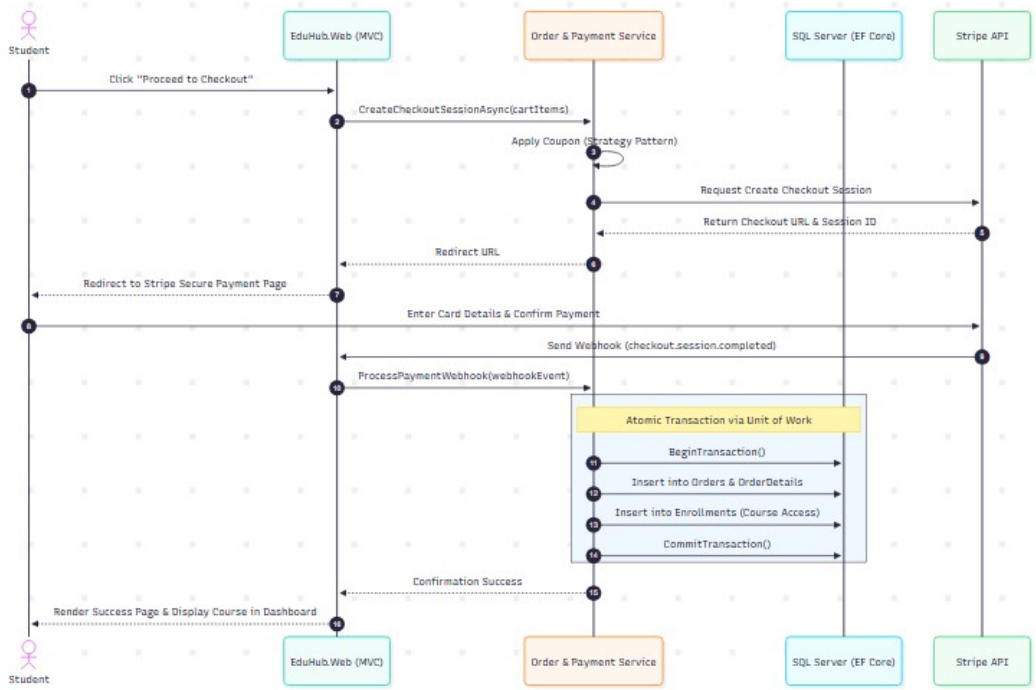
03-Data-Flow-and-System-Behavior

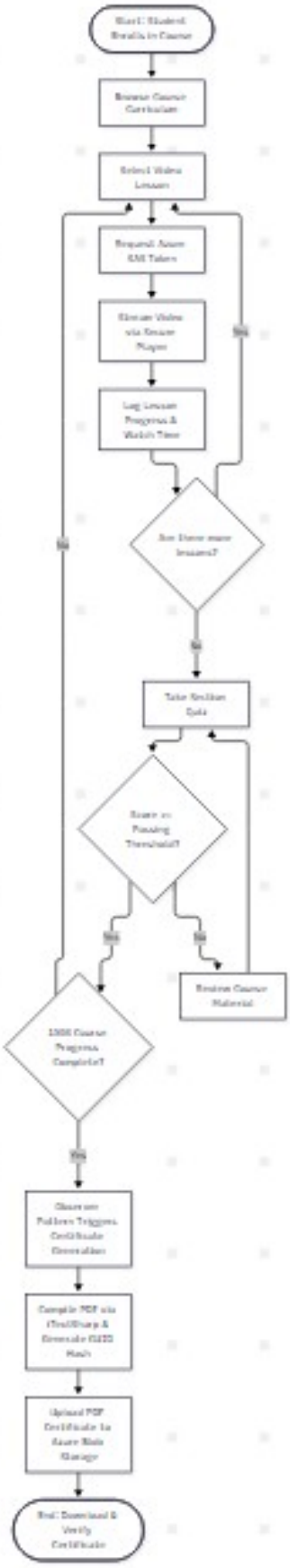
DFD



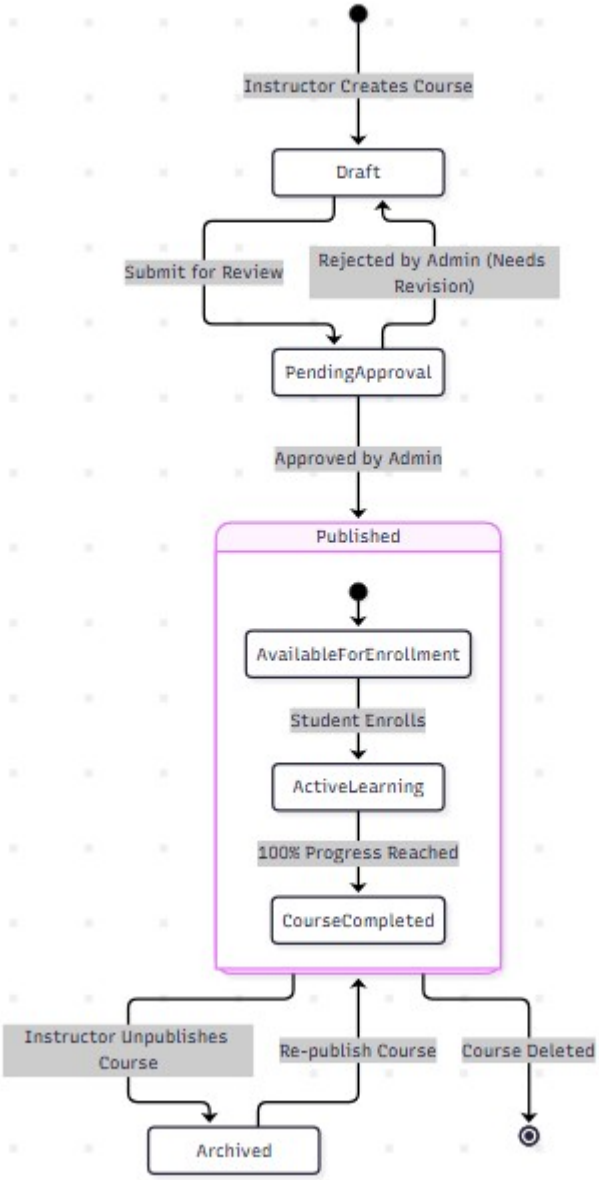
Sequence-Diagrams



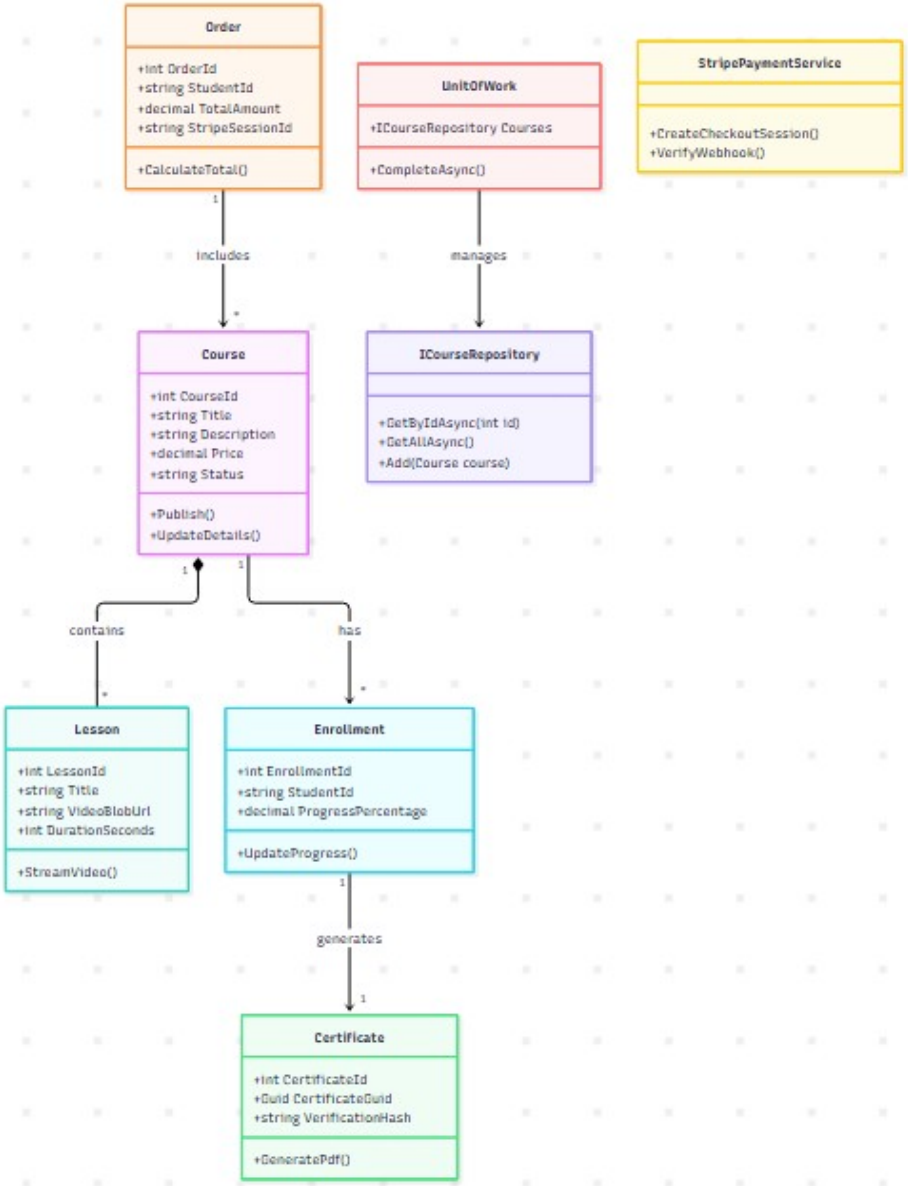
Activity-Diagram


Start: Student Enrolls in Course
Browse Course Curriculum
Select Video Lesson
Request Azure SAS Token
Stream Video via Secure Player
Log Lesson Progress & Watch Time
No
No
Yes
Take Section Quiz
Score >= Passing Threshold?
Yes
No
Review Course Material
100% Course Progress Complete?
Yes
Observer Pattern Triggers Certificate Generation
Compile PDF via iTextSharp & Generate QuestPDF Math
Upload PDF Certificate to Azure Blob Storage
End: Download & Verify Certificate

State-Diagram



Class-Diagram


Order
+int OrderId
+string StudentId
+decimal TotalAmount
+string StripeSessionId
+CalculateTotal()
UnitOfWork
+ICourseRepository Courses
+CompleteAsync()
StripePaymentService
+CreateCheckoutSession()
+VerifyWebhook()
1
includes
*
manages
Course
+int CourseId
+string Title
+string Description
+decimal Price
+string Status
+Publish()
+UpdateDetails()
ICourseRepository
+GetByIdAsync(int id)
+GetAllAsync()
+Add(Course course)
1
1
contains
has
*
*
Lesson
+int LessonId
+string Title
+string VideoBlobUrl
+int DurationSeconds
+StreamVideo()
Enrollment
+int EnrollmentId
+string StudentId
+decimal ProgressPercentage
+UpdateProgress()
1
generates
1
Certificate
+int CertificateId
+Guid CertificateGuid
+string VerificationHash
+GeneratePdf()